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Research Paper:

Generation of Comic Style Chernoff Face with GAN

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This paper proposes a method for generating comic style Chernoff face with generative adversarial network (GAN) as a first step towards the generation of data comics from multi-dimensional data. The proposed method converts Chernoff face into comic style face images based on the combination of CycleGAN and Pix2Pix. Since both Chernoff face graph and comic images do not have enough information for direct conversion, the Chernoff face graphs are converted into photo style face images and then converted into comic images. A questionnaire asking to rank face images according to the specified impressions is conducted to evaluate the proposed method. The result of the questionnaire shows that the proposed method achieved the same level of consistency among answerers' judgments as original Chernoff face. It is also confirmed that the proposed method can express the difference in attribute values with mouth parts.

Keywords: information visualization, comics, Chernoff face, generative adversarial network

1. Introduction

This paper proposes a method for generating comic style Chernoff face as a first step towards the generation of data comics from multi-dimensional data.

Recently, comics and anime are considered effective media for education, advertisement, and medical treatment. Comic computing, which covers various areas such as non-photorealistic rendering (NPR), image processing, database, and machine learning, has been proposed for utilizing the power of comics and anime [1, 2]. Comic computing has been also studied in the field of data visualization. Data comics [3] visualizes data or story as a comic picture. By using data comics, readers' interest in data and story can be aroused by the attractiveness of comic, which is expected to shorten the time for understanding data.

While there are different ways in utilizing comics for communicating with data, this paper aims to represent data as characters. As a means for generating characters from data, this paper focuses on Chernoff face [4], which is one of the popular techniques for visualizing multi-dimensional data with simple face images. By extending the original Chernoff face to comic style one, data can be automatically embedded in data comics as comic characters.

To the best of our knowledge, this paper is the first attempt to convert Chernoff face graph into comic-style images. The challenge is that both Chernoff face graph and comic images, which are drawn with simple lines, do not have enough information to convert original Chernoff face graphs to comic style directly.

To solve it, the proposed method employs two stage conversion. In the first stage, a Chernoff face graph is converted into a photo style face image. In the second stage, the photo style face image is converted into a comic style Chernoff face. As photo style face images have enough features for training Pix2Pix [5], stable conversion is expected to be possible. The paired examples for training Pix2Pix are generated by CycleGAN [6] in similar way as the method of generating comic-style portraits from real face photos [7].

The proposed method is evaluated with questionnaires, which contain 12 Chernoff face graphs and 12 Comic style face graphs. The questionnaire asks the answerers to rank face images according to the specified impressions. The consistency of orders among answerers is evaluated with Kendall's coefficient of concordance [8]. The result shows that a moderate or higher level of agreement was observed among the judgment by the answerers. In addition, the degree of agreement with ground truth is evaluated with normalized discounted cumulative gain (nDCG). The result shows that the proposed method can express the difference in attribute values with mouth parts, although the overall effectiveness of the proposed method against the original Chernoff face depends on image sets.

2. Related Work

2.1. Pix2Pix and CycleGAN

Pix2Pix [5] is an application that converts textures of images from a source domain to a target domain by using conditional adversarial networks. It can generate high quality target domain images by using paired examples for training, even though the source domain images are composed of just stroke lines or noisy images.